GGS 662 AGENTIC GEOAI · RESEARCH PROGRESS

Seeing Through Clouds

Handling clouds in agentic change detection

TEST CASE: HURRICANE IDA 2021 · SPACENET 8 · 0.5 M EO

Justin Rogers · George Mason University · 30 October 2026

Preliminary results on development data · Imagery: Maxar via SpaceNet 8



Clouds hide what our change detector must see

The system. We are building an agentic change detection system: a vision foundation model finds where two satellite images differ, an agent describes what changed, and code checks every fact.

ONE DIFFICULTY · CLOUDS IN EO IMAGERY

"Clouds can hide target areas in images taken after disasters. This can lead to labels being inaccurate compared to ground truth due to obscured parts of the imagery. How should we detect clouds in this imagery, handle them, and report them? The goal is results, such as maps of flooded roads, that people can trust."

Expected contribution: a tested rule for what a change map may say under cloud, a measured label gap, and a test of whether a second date can correct thin cloud.

Design choices. Cloud is judged feature by feature, because labels under cloud cannot be trusted; a second date corrects thin cloud, because the before image shows how dark the ground is.

REPORT

What can a flood-road map reliably tell users about roads hidden by cloud?

HANDLE

When only the before image is cloudy, does the after image alone find change better than both images together?

Tested so far: thin cloud on the after image, corrected from the clear before image.

DETECT · THE TEST BED

Can we use synthetic clouds to represent real-life blockages in satellite imagery?

Proposed method: detect, handle, report

GGS 680 · CLASSICAL

1 · Detect

Greyscale masks on each date: bright cores (our rule), the dark channel (He et al. 2011), and its rise from before to after (our rule).

GGS 662 · THE AGENT GATE

2 · Handle, feature by feature

ASSESS	clear sky
CORRECT	thin cloud + clear second date
CANNOT ASSESS	opaque cloud

Correct = two-date de-hazing inside the cloud mask (our rule, after He et al. 2011), then the detector again (result 3). Filled pixels count as not observed.

WHAT THE MAP MAY SAY

3 · Report

Each statement about a hidden feature gets a tier:

RELIABLE

SYNTHETIC ONLY

RANGE ONLY

CHANGE DETECTOR

Random forest (Breiman 2001) on classical, texture (Haralick 1979) and DINOv2 (Oquab et al. 2024) cues: 83.0% on 798 roads (always “flooded”: 55.3%).

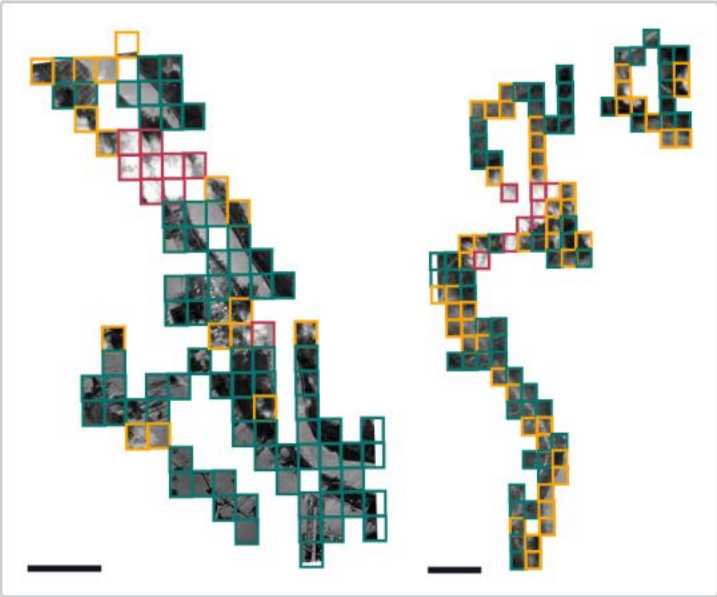
BASELINES AND CHECKS

Tile cloud rule; no correction. Tile-blocked folds (Roberts et al. 2017), McNemar tests (McNemar 1947), one sealed check.

AGENTS PROPOSE, I DECIDE

Question, critic and validation agents draft and check; the handle gate is coded rules. Decided by hand: the problem, the questions, the cloud rule.

Study area and data

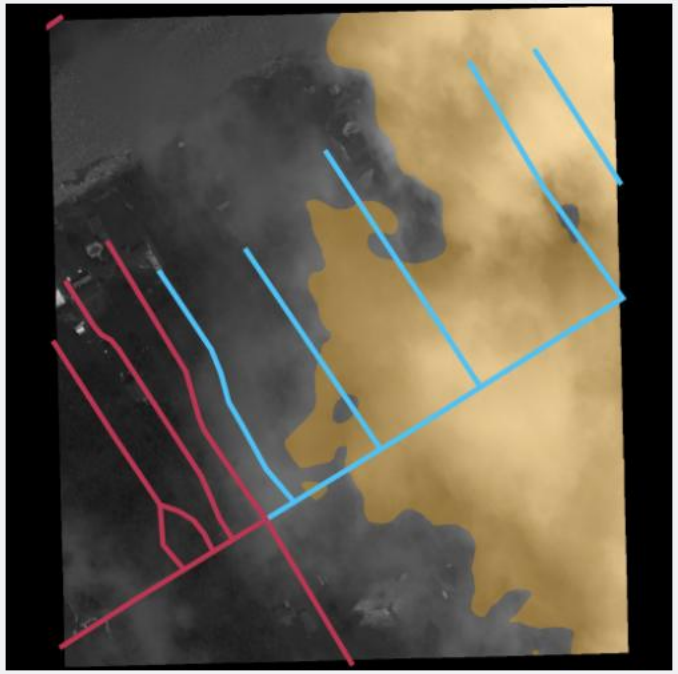


clear · thin cloud or haze · opaque

South (102 tiles) and west (115 tiles) after images, north up, bar 2 km; 29.7°N 90.0°W and 29.8°N 90.1°W · Imagery: Maxar via SpaceNet 8

Imagery	Maxar Open Data before and after, 0.5 m, Hurricane Ida (end of August 2021); used in greyscale
Roads	1,037 OpenStreetMap roads with SpaceNet 8 flood labels (Hänsch et al. 2022), on 217 of the 226 development tiles
Split	186 train · 40 validation · 40 test tiles, plus a 333-tile north negative set; the test tiles, the negative set, Germany and Louisiana-West have never been scored
Cloud labels	HRC_WHU (Li et al. 2019): 150 Google Earth images, 0.5–15 m, expert cloud masks; 120 development, 30 sealed; academic use only
Unit	One road = its OSM line plus a ~5 m band each side; tiles are the fold unit; dates aligned to ~0.6 m
Quality issues	Flood labels are “yes” or missing, never “no” · 17 tiles under opaque cloud · one after image for all development tiles

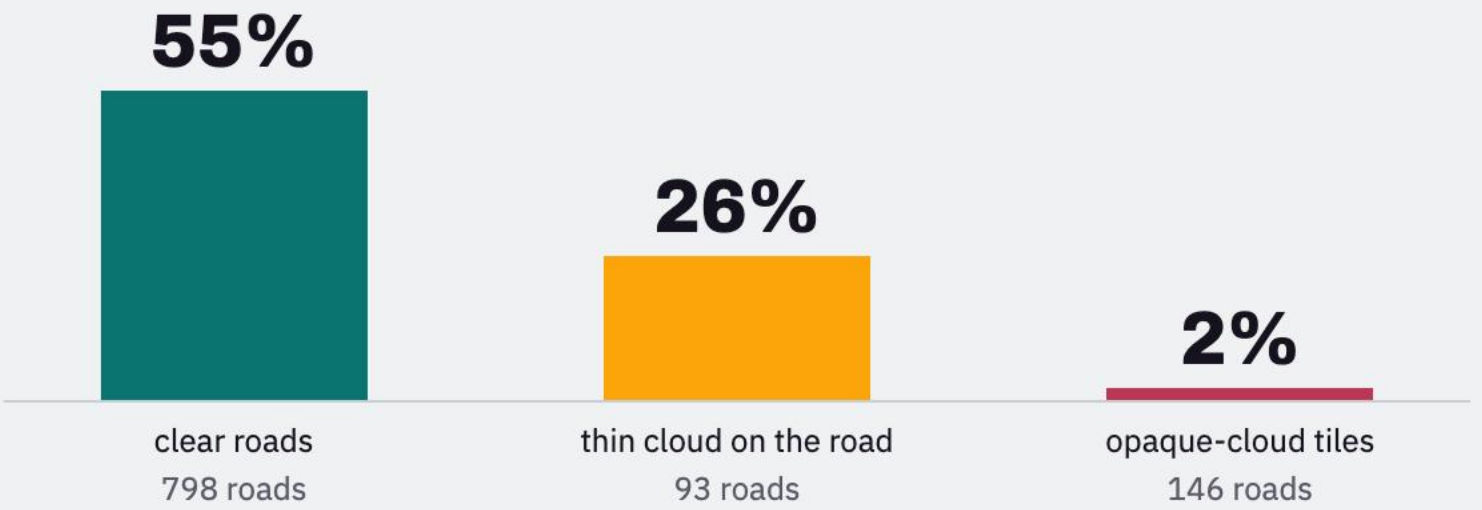
Under cloud, far fewer roads get a flood label



labelled “flooded” thin cloud

no flood label (reads as dry)

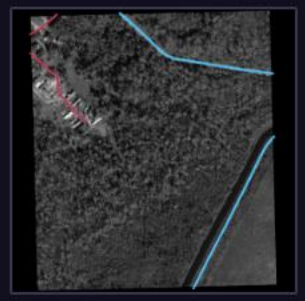
ROADS LABELLED “FLOODED”, BY CLOUD ON THE ROAD



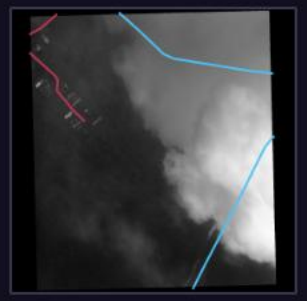
Analysts “only attribute what was visible” (Hänsch et al. 2022), so a flooded road under cloud reads as dry.

Part of the gap is geography: the south scene is dry around its cloudy tiles. Left: an example tile, 2_15_45, after image · Imagery: Maxar via SpaceNet 8

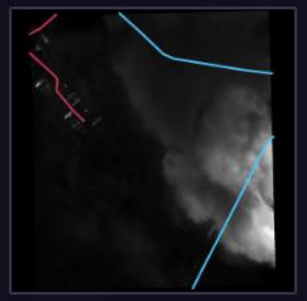
A second image lifts some thin cloud



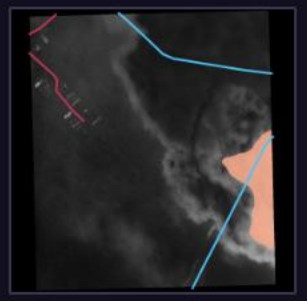
BEFORE · CLEAR



AFTER · CLOUD



SINLGE IMAGE



TWO-DATE

Road picked by a fixed rule: the most metres under thin cloud (481 m, tile 2_23_44).

Red = labelled flooded · blue = no flood label · orange = opaque, cannot assess.

Haze model as used by He et al. (2011), whose single-image method is the baseline; two-date de-hazing is our rule. Imagery: Maxar via SpaceNet 8.

SYNTHETIC HAZE · 94 CLEAR TILES

	none	He 2011	two-date
grey error (0–255)	57	27	37
detail kept (1 = all)	0.82	0.84	1.02
clear altered	0%	85%	0%

He 2011 = single-image baseline.

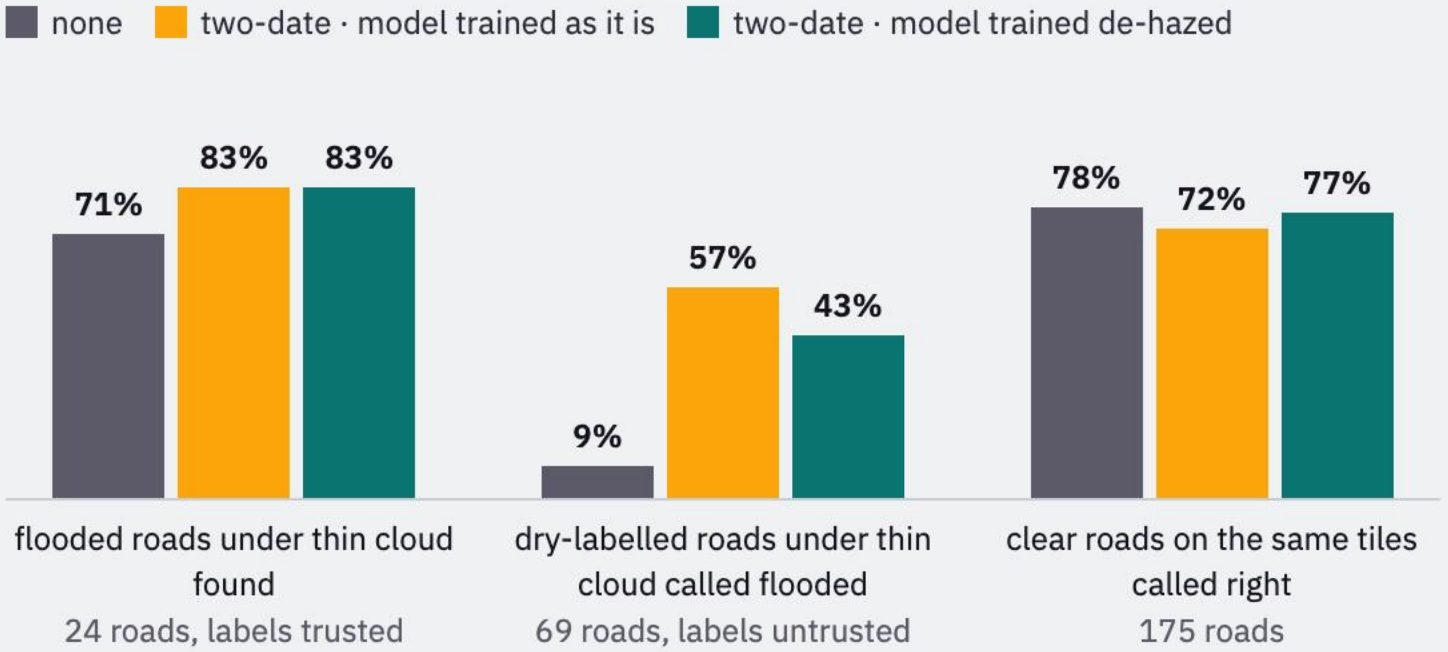
CAVEAT

The synthetic haze follows the two-date model (favours it), and two-date leaves clear ground alone by design; faint haze outside the mask is left alone. Here, little road detail returns.

FLOOD CALLS · EXPLORATORY

Flooded roads under thin cloud found: 20 of 24 two-date, 17 as it is, 21 He 2011. Clear-road accuracy depends on training: result 3.

3 more of 24 flooded roads found, not significant



Production flood model, 5 folds by tile; each run's plan written down first; the 24 flooded roads had been looked at before. Baseline (He 2011), trained as it is like the orange bars: 21 of 24 found, 74% of clear roads right, at least as good as two-date; not yet tried with de-hazed training.

PAIRED TESTS AGAINST NONE

Trained de-hazed: flooded found 4 fixed, 1 broken ($p = 0.38$); clear roads 7 fixed, 9 broken ($p = 0.80$). Trained as it is: clear roads 5 fixed, 15 broken ($p = 0.04$).

CLOUD AS AN EXTRA VARIABLE

No more flooded found (19 of 24, against 20). It cuts calls on cloudy dry-labelled roads (30 → 4), but training on cloudy roads alone does too (7): the labelling rule.

NEXT

Train on de-hazed images and test once on unseen roads.

Feasibility, limits and next steps

Feasible

- Data, masks, de-hazer and the held-out harness all run
- Dark-channel mask: F1 0.65 against 0.41 for the old rule (HRC_WHU)
- Label gap measured: 55%, 26%, 2% labelled flooded (partly geography)
- Each test runs in under an hour
- By semester end: a frozen cloud rule and report tiers, tested once on sealed tiles

Limits

- One event, one sensor, roads only; all real cloud in one after image
- Sealed test tiles sit next to development tiles
- De-hazing: only 24 trusted flooded roads, seen before; cloudy dry-labelled roads flagged
- The method critic's verdict on the plan: revise

Next steps

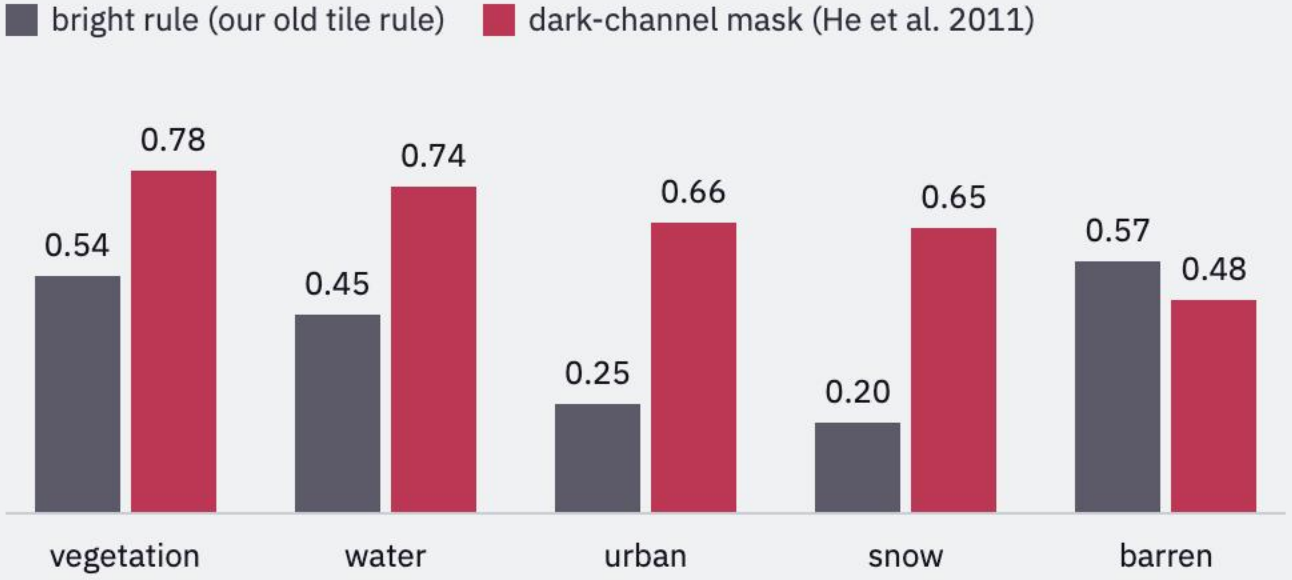
- Pre-register; freeze the cloud generator and de-hazer
- Confirm de-hazed training on unseen roads
- Realism checks and the synthetic test bed
- Before-image cloud
- Report tiers and maps
- One look at the sealed data
- Write the paper

QUESTION FOR DISCUSSION

When a map cannot see the ground, should it say “cannot assess”, show a range, or fill it in from another date?

Cloud masks against human cloud labels

F1 AGAINST EXPERT CLOUD MASKS, BY LAND COVER



0.65

F1, dark-channel mask (precision 0.62, recall 0.68)

0.41

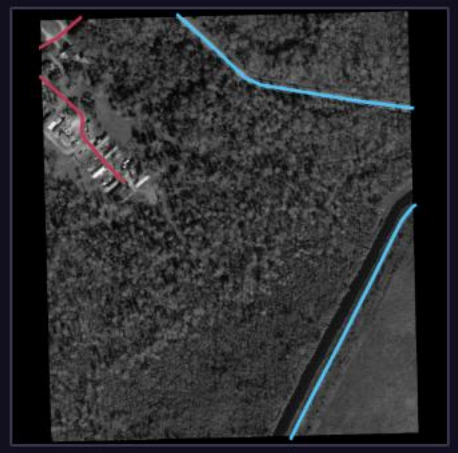
F1, bright rule (precision 0.81, recall 0.28)

120 HRC_WHU development images (Li et al. 2019; 30 sealed). Single-date masks only: the road rule's two-date mask needs a before image.

MASK-BIAS CHECK

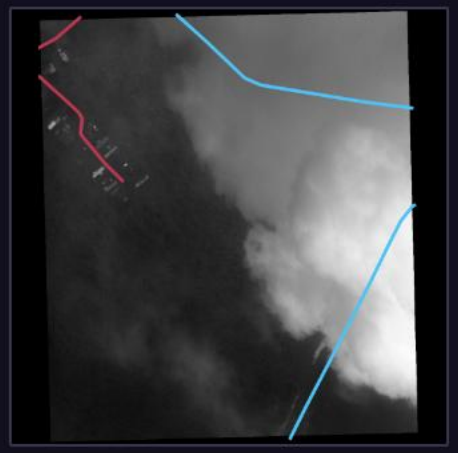
Synthetic haze on 769 clear roads: under thin haze the mask flags flooded and dry roads alike; under thick haze it flags flooded roads more (up to +13 points), not less.

How the two images are layered



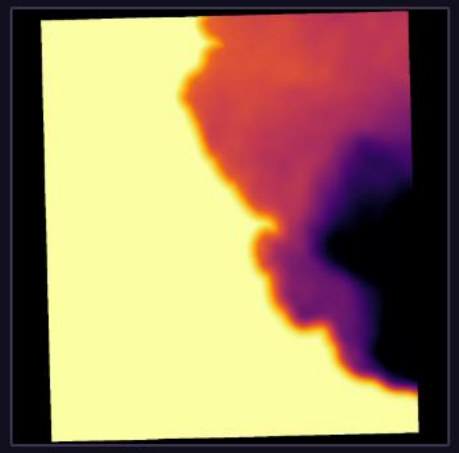
1 · BEFORE

Clear reference: the ground's darkest pixels



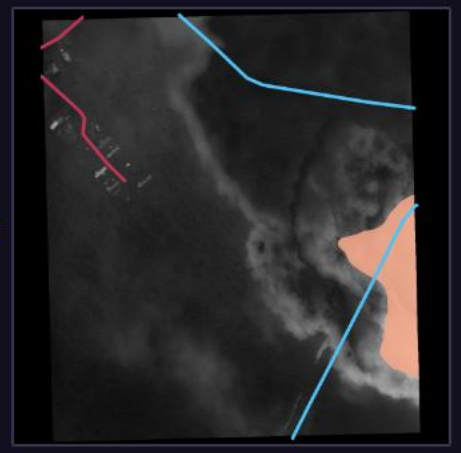
2 · AFTER

Ground dimmed by t, plus airlight A(1 – t)



3 · HAZE t

Inside the cloud mask only; houses stay at 1



4 · CORRECTED

Orange: opaque (t below 0.2), cannot assess

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t = (A − dark after) ÷ (A − tone(dark before)) · corrected = (after − A) ÷ t + A
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Standard haze model and dark channel as used by He et al. (2011). Our rules: the before image's darkest pixels in place of He's zero assumption, the tone curve, a 2,500 m² cloud minimum (so a changed roof is not haze), a 20 m fade past the mask, and t below 0.2 left as opaque.

References, code and AI use

CITED ON THESE SLIDES

Li et al. 2019	HRC_WHU cloud masks · ISPRS J. Photogramm. Remote Sens.
He, Sun & Tang 2011	Dark channel prior · IEEE TPAMI
Oquab et al. 2024	DINOv2 (Meta AI) · TMLR
Haralick 1979	Texture features · Proc. IEEE (abstract checked)
Roberts et al. 2017	Blocked cross-validation · Ecography

BACKGROUND

Canty & Nielsen 2008 (Remote Sensing of Environment) · Portalés-Julià et al. 2023 (Scientific Reports) · Qu, Li & Huang 2026 (Remote Sensing of Environment) · Shen et al. 2015 (IEEE Geoscience and Remote Sensing Magazine) · Bruzzone & Bovolo 2013 (Proc. IEEE)

METHOD ATTRIBUTIONS

Breiman 2001, random forests (Machine Learning) · McNemar 1947, paired proportions (Psychometrika)

Data: SpaceNet 8 flood and road labels, Hänsch et al. 2022 (licences differ: see full references) · HRC_WHU (academic use only)

CODE

github.com/Jrog378/GeoAI-Roads (private; access on request): masks, de-hazing, notebooks

AI USE

Claude Code (Anthropic) agents drafted questions, critiques, notes, code, figures and slides.

Justin chose the problem, the questions and the cloud rule; 28 decisions on AI suggestions are logged in his words (10 rewritten, 6 rejected).

His checks of the haze maps and tiles set a minimum cloud size (houses are not haze) and a label-free tile pick.

A validation agent checked claims against the papers; agent agreement is not validation.

Full references: cited

He, K., Sun, J. & Tang, X. (2011). Single image haze removal using dark channel prior. *IEEE TPAMI* 33(12), 2341–2353. doi:10.1109/TPAMI.2010.168

Li, Z., Shen, H., Cheng, Q., et al. (2019). Deep learning based cloud detection for medium and high resolution remote sensing images of different sensors. *ISPRS J. Photogramm. Remote Sens.* 150, 197–212. doi:10.1016/j.isprsjprs.2019.02.017

Oquab, M., Darcet, T., Moutakanni, T., et al. (2024). DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*. openreview.net/forum?id=a68SUt6zFt

Haralick, R. M. (1979). Statistical and structural approaches to texture. *Proc. IEEE* 67(5), 786–804. doi:10.1109/PROC.1979.11328

Roberts, D. R., Bahn, V., Ciuti, S., et al. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography* 40(8), 913–929. doi:10.1111/ecog.02881

Hänsch, R., Arndt, J., Lunga, D., et al. (2022). SpaceNet 8 – The detection of flooded roads and buildings. *CVPR Workshops*, 1471–1479. doi:10.1109/CVPRW56347.2022.00153

Breiman, L. (2001). Random forests. *Machine Learning* 45(1), 5–32. doi:10.1023/A:1010933404324

McNemar, Q. (1947). Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika* 12(2), 153–157. doi:10.1007/BF02295996

Checked against Crossref, TMLR and the CVF open-access copy (evidence 08). Breiman and McNemar are method attributions.

Full references: background

Canty, M. J. & Nielsen, A. A. (2008). Automatic radiometric normalization of multitemporal satellite imagery with the iteratively re-weighted MAD transformation. *Remote Sens. Environ.* 112(3), 1025–1036. doi:10.1016/j.rse.2007.07.013

Portalés-Julià, E., Mateo-García, G., Purcell, C., et al. (2023). Global flood extent segmentation in optical satellite images. *Scientific Reports* 13, 20316. doi:10.1038/s41598-023-47595-7

Qu, Y., Li, J. & Huang, X. (2026). AnyTime-CD: Self-supervised change detection in arbitrary-length dense time series of high-resolution remote sensing images. *Remote Sens. Environ.* 334, 115239. doi:10.1016/j.rse.2026.115239

Shen, H., Li, X., Cheng, Q., et al. (2015). Missing information reconstruction of remote sensing data: A technical review. *IEEE Geosci. Remote Sens. Mag.* 3(3), 61–85. doi:10.1109/MGRS.2015.2441912

Bruzzone, L. & Bovolo, F. (2013). A novel framework for the design of change-detection systems for very-high-resolution remote sensing images. *Proc. IEEE* 101(3), 609–630. doi:10.1109/JPROC.2012.2197169

Data: SpaceNet 8 imagery from the Maxar Open Data Program; the paper states a Creative Commons 4.0 non-commercial licence, the SpaceNet site CC BY-SA 4.0. HRC_WHU: academic use only.

What is still open

BACKEND · CODE AND RUNS

RUNNING	Road finding under cloud, scored against the OSM roads (as it is vs de-hazed)
TO FIX	Cloud on the before date: the detector fires on 29 clear development tiles
TO FIX	Faint-haze gate: no setting kept clear ground within the 1% limit
TO FIX	De-hazer nudges a corner of cloud-free tiles; fix with a test
TO BUILD	The cloud code as its own library (GeoAI-Clouds), used by the roads repo
TO BUILD	The cloud mask in GeoAI-Descriptions, so cloud reads as cannot assess
TO BUILD	Stage notebooks that rerun these tests from the cached features

DECISIONS AND ACCESS NEEDED

DECIDE	Go-ahead for the one look at the sealed tiles
DECIDE	May synthetic haze count as truth for training?
DECIDE	The definition of change, and the cut-offs for the report tiers
DECIDE	The journal to aim the paper at
ACCESS	SpaceNet 8 licence: paper and site differ; Haralick 1979 full text needs a GMU login